Parity obstructions for the conditioning of sign matrices

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Status of this manuscript: Independently checked partial result; the exact exponent remains open.
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The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently checked the complete argument against the exact target; its detailed review records **PASS** for the partial bound only. The mathematical sections below are unchanged from the reviewed draft. This is independent agent verification, not external human peer review or a formal proof certificate. The submission record preserves the original manuscript and verification history.

Target and result

The repository defines

$$h(n) = \min_{A\in\{-1,1\}^{n\times n}} \kappa_2(A),$$

where singular matrices have infinite condition number, and asks for the exact value

$$\alpha_* = \sup\{\alpha \geq 0 : \exists C > 0\ \forall n \geq 1,\ h(n) - 1 \leq Cn^{-\alpha}\}. \tag{1}$$

Its checked formulation records $17/92 \leq \alpha_* \leq 1$.[1] This note improves the upper bound in that displayed interval, without resolving the exact-value question.

Theorem 1 (odd-order obstruction). If n is odd, every nonsingular sign matrix $A \in \{-1,1\}^{n\times n}$ satisfies

$$\kappa_2(A) \geq \sqrt{1 + \frac{n-1}{n^2}} + \frac{\sqrt{n-1}}{n}. \tag{2}$$

Consequently,

$$h(n) - 1 \geq \frac{\sqrt{n-1}}{n} \quad (n \text{ odd}), \qquad \boxed{\alpha_* \leq \frac{1}{2}}. \tag{3}$$

A variance bound in terms of the condition number

Let $G = A^TA$, let m, M be its smallest and largest eigenvalues, and put $t = M/m = \kappa_2(A)^2$. Since every diagonal entry of G equals n, its eigenvalues $\lambda_1, \dots, \lambda_n$ have mean n. Define their variance

$$v = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i - n)^2 = \frac{1}{n}\operatorname{tr}(G - nI)^2. \tag{4}$$

Summing the nonnegative quantities $(M - \lambda_i)(\lambda_i - m)$ yields

$$v \leq (M - n)(n - m) = (tm - n)(n - m).$$

As a quadratic function of m, the last expression is at most its unrestricted maximum,

$$v \leq \frac{n^2(t-1)^2}{4t}. \tag{5}$$

[1] *Open Problems in Numerical Linear Algebra*, IS-05, "The optimal decay exponent for the conditioning of sign matrices," checked 11 September 2026; canonical entry.

Writing $\kappa = \kappa_2(A) \geq 1$ and taking square roots gives

$$\kappa - \kappa^{-1} \geq \frac{2\sqrt{v}}{n}, \qquad \kappa \geq \sqrt{1 + \frac{v}{n^2}} + \frac{\sqrt{v}}{n}. \tag{6}$$

This derivation also covers $t = 1$; then (5) forces $v = 0$.

Parity supplies the necessary variance

For odd n, the inner product of two sign vectors of length n is an odd integer, so every off-diagonal entry g_{ij} of G has $g_{ij}^2 \geq 1$. Therefore

$$v = \frac{1}{n} \sum_{i \neq j} g_{ij}^2 \geq n - 1. \tag{7}$$

Substituting (7) into (6) proves (2), and hence the first part of (3).

Suppose some $\alpha > 1/2$ were admissible in (1). For every odd n,

$$C \geq n^{\alpha}(h(n) - 1) \geq n^{\alpha - 1}\sqrt{n - 1}.$$

The right-hand side diverges along the odd integers, a contradiction. This proves the upper bound on α_*. Singular matrices cause no exception: their infinite condition numbers satisfy the lower bounds automatically and do not improve the minimum.

An additional bound for orders equal to two modulo four

Theorem 2. If $n \equiv 2 \pmod 4$, every nonsingular sign matrix of order n satisfies

$$\kappa_2(A) \geq \sqrt{1 + \frac{2(n-2)}{n^2}} + \frac{\sqrt{2(n-2)}}{n}. \tag{8}$$

Proof. Form a graph whose vertices are the columns of A, with an edge for each pair of orthogonal columns. This graph is triangle-free. Indeed, if three sign columns were pairwise orthogonal, multiply their rows by signs to make the first column all ones. The second has $n/2$ plus entries and $n/2$ minus entries. Orthogonality of the third to the first two forces its sum on each of these two halves to be zero. Each half must consequently have even size, implying $4 \mid n$, a contradiction.

A triangle-free graph on n vertices has at most $n^2/4$ edges. To see this directly, for every edge uv its endpoint degrees satisfy $d_u + d_v \leq n$. Summing over all e edges gives $\sum_v d_v^2 \leq ne$, whereas Cauchy–Schwarz gives $\sum_v d_v^2 \geq (2e)^2/n$. Thus $e \leq n^2/4$ (including $e = 0$).

At least

$$\frac{n(n-1)}{2} - \frac{n^2}{4} = \frac{n(n-2)}{4}$$

unordered column pairs therefore have nonzero inner products. These inner products are even integers and have square at least four. Counting both orders in (4) gives

$$v \geq \frac{2}{n}\, 4\, \frac{n(n-2)}{4} = 2(n-2).$$

Equation (6) now proves (8).

What remains open

Combining Theorem 1 with the lower bound quoted by the repository gives the proposed updated interval

$$\frac{17}{92} \leq \alpha_* \leq \frac{1}{2}.$$

No construction establishing the matching exponent $1/2$ is supplied here. Theorems 1 and 2 are therefore partial progress, not a solution to the exact-value target. The repository submission records **Partially resolved**, with the exact exponent still open.

The verification script checks the finite-dimensional inequalities on small sign matrices. The all-orders proof is (4)–(8), not the finite tests. Whether these elementary bounds have appeared elsewhere remains to be checked before any novelty claim.

Sources

1. *Open Problems in Numerical Linear Algebra*, IS-05, canonical statement, checked 11 September 2026.
2. B. Alexeev, J. Jasper, and D. G. Mixon, *Asymptotically optimal approximate Hadamard matrices*, arXiv:2511.14653v1, §6, Problem 11; the primary reference identified by the repository for the exponent question and its quoted bounds.
3. The later arXiv version 2, dated 18 August 2026, retains the construction exponent below $17/92$ and the earlier obstruction in §5, Problem 11. The independent review checked this current source; the new upper-bound proof above is unchanged.